JavaScript

Introduction to Software Systems

Web Dev 101

01

JavaScript Basics

The brain and muscle of the web. If HTML is the skeleton and CSS is the skin, JavaScript is what allows the page to move, think, and react.

Adding JS & Variables

Adding JavaScript

Always link your JS file right before the closing body tag inside your HTML.

```
src="app.js">
```

Variables

Where we store and update our data.

```
// app.js
let searchCount = 0;
// 'let' can be changed later

const siteName = "Google";
// 'const' cannot change (constant)
```

Data Types & Logic

Data Types

JS understands different kinds of data.

```
let query = "web dev"; // String
let results = 104000;  // Number
let isDark = false;    // Boolean
```

Arrays

Lists of data ordered by index.

```
let history = ["html", "css"];
console.log(history[0]);
// Prints: "html"
```

Logic (If/Else)

Making decisions in code.

```
if (query === "") {
  alert("Empty!");
} else {
  search(query);
}
```

02

Functions

Reusable blocks of code. Instead of writing the same logic over and over, we package it into a command we can call anytime.

Anatomy of a Function

- ### Write Once, Run Anywhere

  Package your code for later use. This keeps your scripts clean and modular.

- ### Parameters

  Functions can accept input values (parameters) to make them dynamic and adaptable.

- ### Return Values

  A function can process data and send a result back using the `return` keyword.

```
function greetUser(name) {
  return "Hello, " + name + "!";
}

const msg = greetUser("Student");
```

Theme Demo

```
function findSequence(goal) {
  function find(start, history) {
    if (start == goal)
      return history;
    else if (start > goal)
      return null;
    else
      return find(start + 5, "(" + history + " + 5)") ||
             find(start * 3, "(" + history + " * 3)");
  }
  return find(1, "1");
}
```

Select a theme: mbo

03

Events & The DOM

Remember the DOM from Lab 6? Now we use JavaScript to actively listen for user actions and change the page in real-time.

Listening for Actions

1. Select & Listen

Grab the element from the DOM and attach an Event Listener.

```
const searchInput =
document.querySelector("input");

// Listen for "input" (when user types)
searchInput.addEventListener("input",
function() {

  // 2. Code runs on input
  console.log("The user is typing ... ");

});
```

2. React & Update

Read the user's action and update the DOM accordingly.

```
const heading =
document.querySelector("h1");

// Inside the event listener function:

// Read what they typed
let currentText = searchInput.value;

// Change the DOM based on input
heading.textContent = currentText;
```

04

In-Class Assignment

Build a Live Character Counter & Validator.

Take your Google Search Clone from Lab 6 and make the search bar smart.

What You'll Build

1. Page structure

Add a `0/50 characters` below your search input.

2. JavaScript Selection

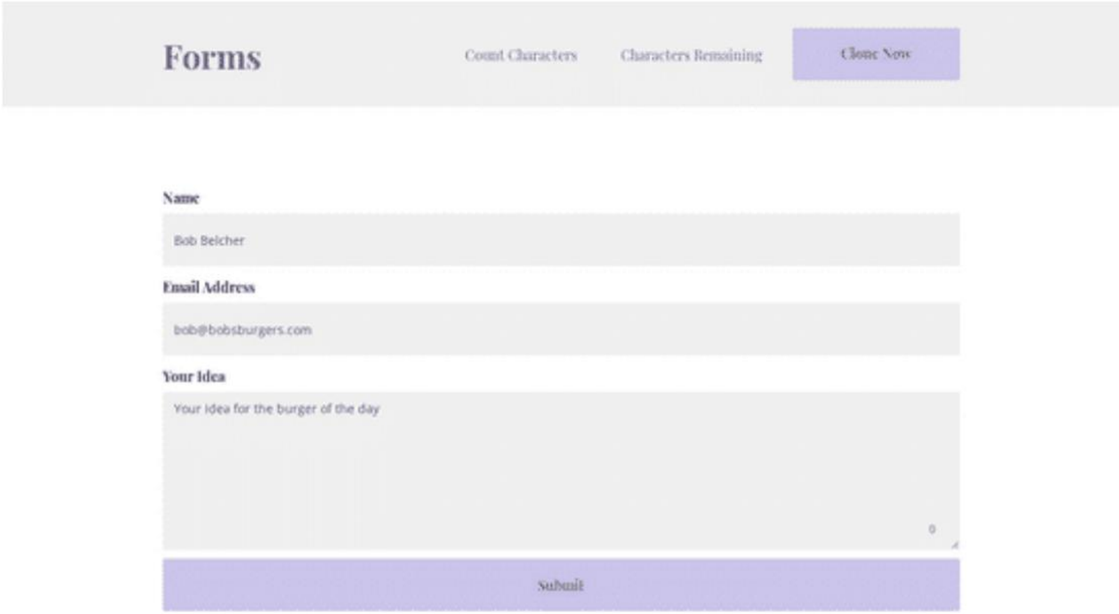
Select the input field, the character count paragraph, and the submit button using `document.querySelector()`.

3. Live Counting

Listen for the "input" event. As the user types, update the paragraph to show the current character length.

4. Form Validation

If the character count is 0, the "Google Search" button should be disabled. If > 0, the button should be clickable.



Tips & Hints

String Length

Get the length of text easily.

```
const txt = "Hello";
console.log(txt.length);
// Outputs: 5
```

Reading Values

Get what the user typed.

```
const inp = document.
  querySelector("input");

let len = inp.value.length;
```

Disable Buttons

Turn buttons off using JS.

```
const btn = document.
  querySelector("button");

btn.disabled = true;
```

Key Takeaways

Variables & Data

Use `let` for variables that change, and `const` for ones that don't. JS handles strings, numbers, booleans, and arrays easily.

Functions

Functions package logic into reusable blocks. Write once, run anywhere!

The DOM is your playground

Use `document.querySelector()` to grab elements, and modify them using `.textContent`, `.style`, or `.classList`.

Event Listeners

`addEventListener` is the bridge between the user and your code. It makes your static HTML come alive in real-time.

Open your editor.

The best way to learn JavaScript is to break things and fix them.

Use your browser's Developer Console (Right Click -> Inspect -> Console) to check for errors!

Good luck with the assignment!